1.
terminal jichi

```
nano /root/soal1.sh
```

```
#!/bin/bash
echo "Konfigurasi jichi dimulai..."

dhclient -v eth0

ip addr flush dev eth1; ip addr add 10.78.1.1/24 dev eth1; ip link
set eth1 up
ip addr flush dev eth2; ip addr add 10.78.2.1/24 dev eth2; ip link
set eth2 up
ip addr flush dev eth3; ip addr add 10.78.3.1/24 dev eth3; ip link
set eth3 up
ip addr flush dev eth4; ip addr add 10.78.4.1/24 dev eth4; ip link
set eth4 up

echo 1 > /proc/sys/net/ipv4/ip_forward
sed -i 's/#net.ipv4.ip_forward=1/net.ipv4.ip_forward=1/g'
/etc/sysctl.conf

apt-get update
apt-get install -y iptables

iptables -t nat -F
iptables -F FORWARD
iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
iptables -P FORWARD ACCEPT

echo "Konfigurasi jichi selesai."
```

```
chmod +x /root/soal1.sh
./root/soal1.sh
```

terminal tengen

```
nano /root/soal1.sh
```

```
#!/bin/bash
echo "Konfigurasi IP Statis tengen..."
ip addr flush dev eth0
ip addr add 10.78.4.10/24 dev eth0
ip link set eth0 up
ip route add default via 10.78.4.1
echo "nameserver 192.168.122.1" > /etc/resolv.conf
```

```
echo "Test koneksi ke internet:"
ping -c 3 8.8.8.8
```

```
chmod +x /root/soal1.sh
./root/soal1.sh
```

terminal network 1

```
nano /root/soal1.sh
```

```
#!/bin/bash
ip addr flush dev eth0
ip addr add 10.78.x.x/24 dev eth0
ip link set eth0 up
ip route add default via 10.78.x.1
echo "nameserver 192.168.122.1" > /etc/resolv.conf

echo "Test koneksi ke internet:"
ping -c 3 8.8.8.8
```

```
chmod +x /root/soal1.sh
./root/soal1.sh
```

terminal network 4

```
nano soal1.sh
```

```
#!/bin/bash
# UBAH ANGKA IP DI BAWAH INI SESUAI NAMA HOST
# Contoh: geto=2, maki=3, panda=4, ..., tengen=10
IP_HOST="10.78.4.2"

echo "Konfigurasi IP Statis: $IP_HOST"
ip addr flush dev eth0
ip addr add $IP_HOST/24 dev eth0
ip link set eth0 up
ip route add default via 10.78.4.1
echo "nameserver 192.168.122.1" > /etc/resolv.conf

echo "Test koneksi ke internet:"
ping -c 3 8.8.8.8
```

2.
terminal tengen

```
nano /root/soal2.sh
```

```
#!/bin/bash
echo "Memulai konfigurasi DHCP Server pada tengen..."

# 1. Update dan instal DHCP Server
apt-get update
apt-get install -y isc-dhcp-server

# 2. Atur Interface DHCP agar mendengarkan di eth0
echo 'INTERFACESv4="eth0"' > /etc/default/isc-dhcp-server

# 3. Tulis ulang isi dhcpd.conf
cat << 'CONF' > /etc/dhcp/dhcpd.conf
# Konfigurasi Global
default-lease-time 600;
max-lease-time 7200;
authoritative;

# Deklarasi Network 4 (Subnet Lokal tengen)
subnet 10.78.4.0 netmask 255.255.255.0 {
}

# Deklarasi Scope Network 2 (Divisi A)
subnet 10.78.2.0 netmask 255.255.255.0 {
    range 10.78.2.10 10.78.2.100;
    option routers 10.78.2.1;
    option domain-name-servers 192.168.122.1;
}

# Deklarasi Scope Network 3 (Divisi B)
subnet 10.78.3.0 netmask 255.255.255.0 {
    range 10.78.3.10 10.78.3.100;
    option routers 10.78.3.1;
    option domain-name-servers 192.168.122.1;
}
CONF

# 4. Restart layanan dan tampilkan status
service isc-dhcp-server restart
echo "Status isc-dhcp-server:"
service isc-dhcp-server status
```

```
chmod +x /root/soal2.sh
./root/soal2.sh
```

3.

terminal jichi

```
nano /root/soal3.sh
```

```
#!/bin/bash
echo "Memulai konfigurasi DHCP Relay pada jichi..."

# 1. Update dan instal isc-dhcp-relay
apt-get update
apt-get install -y isc-dhcp-relay

# 2. Konfigurasi file default relay
echo 'SERVERS="10.78.4.10"' > /etc/default/isc-dhcp-relay
echo 'INTERFACES="eth2 eth3 eth4"' >> /etc/default/isc-dhcp-relay
echo 'OPTIONS=""' >> /etc/default/isc-dhcp-relay

# 3. Restart layanan relay
service isc-dhcp-relay restart

echo "Status isc-dhcp-relay:"
service isc-dhcp-relay status
```

```
chmod +x /root/soal3_jichi.sh
./root/soal3_jichi.sh
```

terminal network 2

```
nano soal3.sh
```

```
#!/bin/bash
# Hapus IP yang mungkin tersisa
ip addr flush dev eth0

# Panggil DHCP client bawaan Alpine (udhcpc)
udhcpc -i eth0

# Cek IP yang didapatkan
ip a show eth0
```

4.

terminal pada **itadori**, **fushiguro**, **zenin**, dan **okkotsu**

```
ip link show eth0
```

terminal tengen

```
nano /root/soal4.sh
```

```
#!/bin/bash
echo "Memulai konfigurasi Multi-Pool DHCP pada tengen..."

cat << 'CONF' > /etc/dhcp/dhcpd.conf
# Konfigurasi Global
default-lease-time 600;
max-lease-time 7200;
authoritative;

# ----------------- KELAS MAC ADDRESS -----------------
# Kelas untuk Tim Dev (itadori, fushiguro)
class "tim-dev" {
    match hardware;
}
# UBAH MAC ADDRESS DI BAWAH INI (biarkan angka 1: di depannya)
subclass "tim-dev" 1:02:42:24:21:1a:00; # Ganti dengan MAC itadori
subclass "tim-dev" 1:02:42:30:24:4c:00; # Ganti dengan MAC
fushiguro

# Kelas untuk Tim Support (zenin, okkotsu)
class "tim-support" {
    match hardware;
}
# UBAH MAC ADDRESS DI BAWAH INI (biarkan angka 1: di depannya)
subclass "tim-support" 1:02:42:28:b9:c0:00; # Ganti dengan MAC
zenin
subclass "tim-support" 1:02:42:d3:fe:e9:00; # Ganti dengan MAC
okkotsu


# ----------------- DEKLARASI SUBNET -----------------
# Deklarasi Network 4 (Subnet Lokal tengen)
subnet 10.78.4.0 netmask 255.255.255.0 {
}

# Deklarasi Scope Network 2 (Divisi A) dengan Multi-Pool
subnet 10.78.2.0 netmask 255.255.255.0 {
    option routers 10.78.2.1;
    option domain-name-servers 192.168.122.1;
```

```
    # Pool 1: Khusus untuk member kelas "tim-dev"
    pool {
        allow members of "tim-dev";
        range 10.78.2.10 10.78.2.35;
    }

    # Pool 2: Khusus untuk member kelas "tim-support"
    pool {
        allow members of "tim-support";
        range 10.78.2.70 10.78.2.95;
    }
}

# Deklarasi Scope Network 3 (Divisi B)
subnet 10.78.3.0 netmask 255.255.255.0 {
    range 10.78.3.10 10.78.3.100;
    option routers 10.78.3.1;
    option domain-name-servers 192.168.122.1;
}
CONF

# Restart layanan
service isc-dhcp-server restart
echo "Status isc-dhcp-server:"
service isc-dhcp-server status
```

```
chmod +x /root/soal4.sh
./root/soal4.sh
```

5.

terminal pada **ino**, **kusakabe**, dan **miwa**

```
ip link show eth0
```

terminal tengen

```
nano /root/soal5.sh
```

```
#!/bin/bash
echo "Memulai konfigurasi Lease Time dan Options pada tengen..."

cat << 'CONF' > /etc/dhcp/dhcpd.conf
# Konfigurasi Global
default-lease-time 600;
max-lease-time 7200;
authoritative;
```

```
# ----------------- KELAS NETWORK 2 (Soal 4) -----------------
class "tim-dev" { match hardware; }
subclass "tim-dev" 1:02:42:24:21:1a:00; # MAC itadori
subclass "tim-dev" 1:02:42:30:24:4c:00; # MAC fushiguro

class "tim-support" { match hardware; }
subclass "tim-support" 1:02:42:28:b9:c0:00; # MAC zenin
subclass "tim-support" 1:02:42:d3:fe:e9:00; # MAC okkotsu

# ----------------- KELAS NETWORK 3 (Soal 5) -----------------
class "short-lease" { match hardware; }
subclass "short-lease" 1:02:42:8a:f1:0f:00; # MAC ino
subclass "short-lease" 1:02:42:a5:de:b3:00; # MAC kusakabe

class "long-lease" { match hardware; }
subclass "long-lease" 1:02:42:80:41:bf:00; # MAC miwa


# ----------------- DEKLARASI SUBNET -----------------
# Deklarasi Network 4
subnet 10.78.4.0 netmask 255.255.255.0 {
}

# Deklarasi Network 2 (Divisi A)
subnet 10.78.2.0 netmask 255.255.255.0 {
    option routers 10.78.2.1;
    option domain-name-servers 192.168.122.1;

    pool {
        allow members of "tim-dev";
        range 10.78.2.10 10.78.2.35;
    }
    pool {
        allow members of "tim-support";
        range 10.78.2.70 10.78.2.95;
    }
}

# Deklarasi Network 3 (Divisi B)
subnet 10.78.3.0 netmask 255.255.255.0 {
    option routers 10.78.3.1;

    # DHCP Options Spesifik Network 3
    option domain-name-servers 10.78.1.2, 10.78.1.3;
    option domain-name "k29.com";

    # Pool 1: Short Lease (30 menit = 1800 detik)
    pool {
        allow members of "short-lease";
```

```
        range 10.78.3.10 10.78.3.89;
        default-lease-time 1800;
        max-lease-time 1800;
    }

    # Pool 2: Long Lease (12 jam = 43200 detik)
    pool {
        allow members of "long-lease";
        range 10.78.3.100 10.78.3.199;
        default-lease-time 43200;
        max-lease-time 43200;
    }
}
CONF

# Restart layanan DHCP Server
service isc-dhcp-server restart
echo "Status isc-dhcp-server:"
service isc-dhcp-server status
```

```
chmod +x /root/soal5_tengen.sh
./root/soal5_tengen.sh
```

terminal network 3

```
ip addr flush dev eth0
udhcpc -i eth0
```

```
cat /etc/resolv.conf
```

6.

terminal kugisaki

```
ip link show eth0
```

terminal tengen

```
nano soal6.sh
```

```
#!/bin/bash
echo "Memulai konfigurasi Lease Time dan Options pada tengen..."

cat << 'CONF' > /etc/dhcp/dhcpd.conf
# Konfigurasi Global
default-lease-time 600;
```

```
max-lease-time 7200;
authoritative;

# ----------------- KELAS NETWORK 2 (Soal 4) -----------------
class "tim-dev" { match hardware; }
subclass "tim-dev" 1:02:42:24:21:1a:00; # MAC itadori
subclass "tim-dev" 1:02:42:30:24:4c:00; # MAC fushiguro

class "tim-support" { match hardware; }
subclass "tim-support" 1:02:42:28:b9:c0:00; # MAC zenin
subclass "tim-support" 1:02:42:d3:fe:e9:00; # MAC okkotsu

# ----------------- KELAS NETWORK 3 (Soal 5) -----------------
class "short-lease" { match hardware; }
subclass "short-lease" 1:02:42:8a:f1:0f:00; # MAC ino
subclass "short-lease" 1:02:42:a5:de:b3:00; # MAC kusakabe

class "long-lease" { match hardware; }
subclass "long-lease" 1:02:42:80:41:bf:00; # MAC miwa


# ----------------- DEKLARASI SUBNET -----------------
# Deklarasi Network 4
subnet 10.78.4.0 netmask 255.255.255.0 {
}

# Deklarasi Network 2 (Divisi A)
subnet 10.78.2.0 netmask 255.255.255.0 {
    option routers 10.78.2.1;
    option domain-name-servers 192.168.122.1;

    pool {
        allow members of "tim-dev";
        range 10.78.2.10 10.78.2.35;
    }
    pool {
        allow members of "tim-support";
        range 10.78.2.70 10.78.2.95;
    }
}

# Deklarasi Network 3 (Divisi B)
subnet 10.78.3.0 netmask 255.255.255.0 {
    option routers 10.78.3.1;

    # DHCP Options Spesifik Network 3
    option domain-name-servers 10.78.1.2, 10.78.1.3;
    option domain-name "k29.com";
```

```
    # Pool 1: Short Lease (30 menit = 1800 detik)
    pool {
        allow members of "short-lease";
        range 10.78.3.10 10.78.3.89;
        default-lease-time 1800;
        max-lease-time 1800;
    }

    # Pool 2: Long Lease (12 jam = 43200 detik)
    pool {
        allow members of "long-lease";
        range 10.78.3.100 10.78.3.199;
        default-lease-time 43200;
        max-lease-time 43200;
    }
}

# Reservasi IP VIP Kugisaki (Soal 6)
host kugisaki {
    hardware ethernet 02:42:8f:4e:be:00; # Ganti dengan MAC
Address Kugisaki
    fixed-address 10.78.3.99;
}
CONF

# Restart layanan DHCP Server
service isc-dhcp-server restart
echo "Status isc-dhcp-server:"
service isc-dhcp-server status
```

terminal kugisaki

```
./soal3.sh
```

7.

terminal itadori, zenin, ino, atau miwa

```
nano soal7.sh
```

```
#!/bin/bash
echo "=== VALIDASI KONFIGURASI JARINGAN CLIENT (SOAL 7) ==="

echo -e "\n1. IP Address, Subnet Mask (/24), dan Broadcast:"
ip a show eth0

echo -e "\n2. Tabel Routing (Default Gateway):"
ip route
```

```
echo -e "\n3. Konfigurasi DNS Server dan Domain:"
cat /etc/resolv.conf

echo -e "\n=== VALIDASI SELESAI ==="
```

terminal tengen

```
nano /root/soal7.sh
```

```
#!/bin/bash
echo "=== VALIDASI DATABASE LEASE DHCP SERVER (SOAL 7) ==="

echo -e "\nMenampilkan daftar IP dinamis yang telah disewakan:"
cat /var/lib/dhcp/dhcpd.leases | grep -E "lease|binding
state|hardware ethernet"

echo -e "\n=== VALIDASI SELESAI ==="
echo "Catatan: IP 10.78.3.99 milik Kugisaki tidak akan muncul di
sini karena menggunakan static lease."
```

8.
terminal tengen

```
nano soal8.sh
```

```
#!/bin/bash
echo "=== PACKET CAPTURE PROTOKOL DORA DHCP (SOAL 8) ==="

# Instal tcpdump jika belum ada
apt-get update
apt-get install -y tcpdump

echo -e "\n[MENUNGGU PAKET DHCP...]"
echo "Buka terminal client (misal: itadori di Divisi A atau ino di
Divisi B)."
echo "Jalankan perintah ini di client: ip addr flush dev eth0 &&
udhcpc -i eth0"
echo -e "Tekan Ctrl+C di sini untuk berhenti jika client sudah
dapat IP.\n"
echo
"-----------------------------------------------------------------
-"

# Menjalankan tcpdump pada port 67 dan 68 (port standar DHCP)
dengan output detail (-vvv)
tcpdump -i eth0 -n -vvv port 67 or port 68
```

```
chmod +x /root/soal8_tengen.sh
./root/soal8_tengen.sh
```

selagi tengen berjalan jalankan di itadori atau client lain

```
ip addr flush dev eth0
udhcpc -i eth0
```

9.
terminal jichi

```
service isc-dhcp-relay stop
```

```
ps aux | grep dhcrelay
```

terminal itadori

```
ip addr flush dev eth0
udhcpc -i eth0
```

terminal jichi

```
service isc-dhcp-relay start
# Atau bisa menggunakan script yang sudah Anda buat sebelumnya:
# ./root/soal3.sh
```

terminal itadori

```
udhcpc -i eth0
ip a show eth0
ip route
```

10.
terminal tengen

```
# Menambahkan perintah start ke .bashrc
echo "service isc-dhcp-server start" >> /root/.bashrc

# (Opsional) Jika node Anda mendukung systemd, jalankan juga:
systemctl enable isc-dhcp-server
```

terminal jinchi

```
# Memanggil script konfigurasi IP dan menyalakan relay otomatis
echo "/root/soal1.sh" >> /root/.bashrc
echo "service isc-dhcp-relay start" >> /root/.bashrc
```

11.
terminal utahime

```
nano /root/soal11.sh
```

```
#!/bin/bash
echo "Konfigurasi DNS Master (utahime)..."

apt-get update
apt-get install -y bind9 bind9utils

# Konfigurasi zona master dan allow-transfer ke IP shoko
cat << 'EOF' > /etc/bind/named.conf.local
zone "jujutsu.k29.com" {
    type master;
    file "/etc/bind/jujutsu/jujutsu.k29.com";
    allow-transfer { 10.78.1.3; };
};
EOF

# Membuat direktori dan file zona
mkdir -p /etc/bind/jujutsu
cat << 'EOF' > /etc/bind/jujutsu/jujutsu.k29.com
$TTL    604800
@       IN      SOA     utahime.jujutsu.k29.com.
root.jujutsu.k29.com. (
                              2026101501 ; Serial
                              604800     ; Refresh
                              86400      ; Retry
                              2419200    ; Expire
                              604800 )   ; Negative Cache TTL
;
@       IN      NS      utahime.jujutsu.k29.com.
@       IN      NS      shoko.jujutsu.k29.com.
utahime IN      A       10.78.1.2
shoko   IN      A       10.78.1.3
EOF

# Menjalankan layanan menggunakan nama 'named'
service named restart

echo "Status Layanan DNS Master:"
ps aux | grep named | grep -v grep
```

```
chmod +x /root/soal11_master.sh
./root/soal11_master.sh
```

terminal shoko

```
nano /root/soal11.sh
```

```
#!/bin/bash
echo "Konfigurasi DNS Slave (shoko)..."

apt-get update
apt-get install -y bind9 bind9utils

# Konfigurasi zona slave yang menunjuk ke IP utahime
cat << 'EOF' > /etc/bind/named.conf.local
zone "jujutsu.k29.com" {
    type slave;
    masters { 10.78.1.2; };
    file "/var/lib/bind/jujutsu.k29.com";
};
EOF

# Menjalankan layanan menggunakan nama 'named'
service named restart

echo "Status Layanan DNS Slave:"
ps aux | grep named | grep -v grep
```

```
chmod +x /root/soal11_slave.sh
./root/soal11_slave.sh
```

terminal utahime

```
nano /etc/bind/jujutsu/jujutsu.k29.com
```

Ubah angka Serial dari 2026101501 menjadi 2026101502

```
service named restart
# atau service bind9 reload
```

terminal shoko

```
cat /var/lib/bind/jujutsu.k29.com
```

terminal itadori

```
# Instal alat query (jika belum ada)
apk update
apk add bind-tools

# Screenshot 1: Uji Master
dig @10.78.1.2 NS jujutsu.k29.com
```

```
# Screenshot 2: Uji Slave
dig @10.78.1.3 NS jujutsu.k29.com
```

12.
terminal utahime

```
nano /root/soal12.sh
```

```
#!/bin/bash
echo "Menambahkan A Record, CNAME, PTR, dan Forwarder di
utahime..."

# Asumsi IP Gojo (UBAH JIKA BERBEDA)
IP_GOJO="10.78.1.4"
OCTET_TERAKHIR="4"  # Angka terakhir dari IP Gojo (dari 10.78.1.4)

# 1. Menambahkan Forwarder ke 192.168.122.1
cat << 'EOF' > /etc/bind/named.conf.options
options {
    directory "/var/cache/bind";

    forwarders {
        192.168.122.1;
    };

    allow-query { any; };
    dnssec-validation no;
    listen-on-v6 { any; };
};
EOF

# 2. Update File Zona Forward (Menambahkan A record Gojo & CNAME
www)
# Perhatikan Serial dinaikkan menjadi 2026101503
cat << 'EOF' > /etc/bind/jujutsu/jujutsu.k29.com
$TTL    604800
@       IN      SOA     utahime.jujutsu.k29.com.
root.jujutsu.k29.com. (
                              2026101503 ; Serial
                              604800     ; Refresh
                              86400      ; Retry
                              2419200    ; Expire
                              604800 )   ; Negative Cache TTL
;
@       IN      NS      utahime.jujutsu.k29.com.
@       IN      NS      shoko.jujutsu.k29.com.
@       IN      A       10.78.1.4
```

```
www     IN      CNAME   jujutsu.k29.com.
gojo    IN      A       10.78.1.4
utahime IN      A       10.78.1.2
shoko   IN      A       10.78.1.3
EOF

# 3. Mendaftarkan Zona Reverse di named.conf.local
# Jika IP Gojo 10.78.1.4, maka zona reverse-nya adalah
1.78.10.in-addr.arpa
cat << 'EOF' > /etc/bind/named.conf.local
zone "jujutsu.k29.com" {
    type master;
    file "/etc/bind/jujutsu/jujutsu.k29.com";
    allow-transfer { 10.78.1.3; };
};

zone "1.78.10.in-addr.arpa" {
    type master;
    file "/etc/bind/jujutsu/10.78.1.rev";
};
EOF

# 4. Membuat File Zona Reverse (PTR)
cat << 'EOF' > /etc/bind/jujutsu/10.78.1.rev
$TTL    604800
@       IN      SOA     utahime.jujutsu.k29.com.
root.jujutsu.k29.com. (
                              2026101501 ; Serial
                              604800     ; Refresh
                              86400      ; Retry
                              2419200    ; Expire
                              604800 )   ; Negative Cache TTL
;
@       IN      NS      utahime.jujutsu.k29.com.
@       IN      NS      shoko.jujutsu.k29.com.
4       IN      PTR     jujutsu.k29.com.
EOF

# Restart Layanan BIND9 (named)
service named restart

echo "Konfigurasi Soal 12 Selesai. Status BIND9:"
service named status
```

```
chmod +x /root/soal12.sh
./root/soal12.sh
```

terminal itadori

a. menguji CNAMe

```
dig @10.78.1.2 www.jujutsu.k29.com
```

b. menguji ptr

```
dig @10.78.1.2 -x 10.78.1.4
```

c. Menguji Forwarder

```
dig @10.78.1.2 google.com
```

d.

```
ping -c 3 google.com
```

ip
maki: 10.78.4.12
panda: 10.78.4.13
nanami: 10.78.4.14
todo: 10.78.4.15
kirara: 10.78.4.16
gojo: 10.78.1.4

SOAL 13

Konfigurasi DB MASTER **yaga**

```
cat << 'EOF' > /root/setup_yaga.sh
#!/bin/bash

# 1. Pastikan nameserver awal aktif untuk instalasi paket
echo "nameserver 192.168.122.1" > /etc/resolv.conf

# 2. Instalasi MariaDB Server
apt-get update
apt-get install -y mariadb-server

# 3. Siapkan direktori kerja dan izin akses log (Mencegah error
socket/index)
mkdir -p /run/mysqld /var/log/mysql /var/lib/mysql
chown -R mysql:mysql /run/mysqld /var/log/mysql /var/lib/mysql
chmod -R 755 /run/mysqld /var/log/mysql

# 4. Konfigurasi 50-server.cnf (Buka bind-address & aktifkan
Binary Logging)
cat << 'CNF' > /etc/mysql/mariadb.conf.d/50-server.cnf
[server]
[mysqld]
user                    = mysql
pid-file                = /run/mysqld/mysqld.pid
socket                  = /run/mysqld/mysqld.sock
port                    = 3306
basedir                 = /usr
datadir                 = /var/lib/mysql
tmpdir                  = /tmp
lc-messages-dir         = /usr/share/mysql
lc-messages             = en_US
skip-external-locking

# Mengizinkan akses remote dari Network 4
```

```
bind-address            = 0.0.0.0

# Konfigurasi Master & Binary Logging
server-id               = 1
log_bin                 = /var/log/mysql/mariadb-bin
binlog_do_db            = db_shibuya_k29

expire_logs_days        = 10
character-set-server    = utf8mb4
collation-server        = utf8mb4_general_ci

[embedded]
[mariadb]
[mariadb-10.5]
CNF

# 5. Restart layanan MariaDB untuk memuat konfigurasi baru
service mariadb restart

# 6. Konfigurasi User, Hak Akses Network 4, Database, dan Tabel
PREFIX_IP="10.78"

mariadb -u root << SQL
-- Buat user khusus untuk replikasi Slave (inumaki)
CREATE USER IF NOT EXISTS 'repl_user'@'${PREFIX_IP}.4.%'
IDENTIFIED BY 'repl_pass';
GRANT REPLICATION SLAVE ON *.* TO 'repl_user'@'${PREFIX_IP}.4.%';

-- Berikan hak akses remote bagi seluruh node di Network 4 (Worker
API, dll)
CREATE USER IF NOT EXISTS 'shibuya_app'@'${PREFIX_IP}.4.%'
IDENTIFIED BY 'shibuya_pass';
GRANT ALL PRIVILEGES ON db_shibuya_k29.* TO
'shibuya_app'@'${PREFIX_IP}.4.%';
FLUSH PRIVILEGES;

-- Buat database dan tabel request_logs
CREATE DATABASE IF NOT EXISTS db_shibuya_k29;
USE db_shibuya_k29;

CREATE TABLE IF NOT EXISTS request_logs (
    id INT AUTO_INCREMENT PRIMARY KEY,
    worker_name VARCHAR(50),
    client_ip VARCHAR(45) NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
SQL

# 7. Tampilkan posisi Binary Log untuk dicatat ke Slave
```

```
echo "=== CATAT FILE DAN POSITION DI BAWAH INI UNTUK INUMAKI ==="
mariadb -u root -e "SHOW MASTER STATUS\G"
EOF

chmod +x /root/setup_yaga.sh
bash /root/setup_yaga.sh


chmod +x /root/setup_yaga.sh
bash /root/setup_yaga.sh
```

Konfigurasi DB SLAVE **inumaki**

```
cat << 'EOF' > /root/setup_inumaki.sh
#!/bin/bash

# 1. Pastikan nameserver awal aktif untuk instalasi paket
echo "nameserver 192.168.122.1" > /etc/resolv.conf

# 2. Instalasi MariaDB Server
apt-get update
apt-get install -y mariadb-server

# 3. Konfigurasi 50-server.cnf (Tetapkan server-id = 2)
cat << 'CNF' > /etc/mysql/mariadb.conf.d/50-server.cnf
[server]
[mysqld]
user                    = mysql
pid-file                = /run/mysqld/mysqld.pid
socket                  = /run/mysqld/mysqld.sock
port                    = 3306
basedir                 = /usr
datadir                 = /var/lib/mysql
tmpdir                  = /tmp
lc-messages-dir         = /usr/share/mysql
lc-messages             = en_US
skip-external-locking

bind-address            = 0.0.0.0

# Konfigurasi Slave (server-id wajib berbeda dari yaga)
server-id               = 2
relay-log               = /var/log/mysql/mariadb-relay-bin
log_bin                 = /var/log/mysql/mariadb-bin
binlog_do_db            = db_shibuya_k29
read_only               = 1
```

```
character-set-server    = utf8mb4
collation-server        = utf8mb4_general_ci
CNF

# 4. Restart layanan MariaDB
service mariadb restart

# 5. Siapkan skema dasar dan hubungkan ke Master (yaga)
# SESUAIKAN 3 variabel di bawah ini dengan topologi & output SHOW
MASTER STATUS di yaga!
IP_YAGA="10.29.4.X"              # Ganti dengan IP statis node
yaga sesuai tabel kelompok
LOG_FILE="mariadb-bin.000001"    # Sesuaikan dengan output File di
yaga
LOG_POS=654                      # Sesuaikan dengan angka Position
di yaga (tanpa tanda kutip)

mariadb -u root << SQL
STOP SLAVE;

-- Siapkan struktur database dan tabel agar sinkron dengan posisi
terakhir yaga
CREATE DATABASE IF NOT EXISTS db_shibuya_k29;
USE db_shibuya_k29;

CREATE TABLE IF NOT EXISTS request_logs (
    id INT AUTO_INCREMENT PRIMARY KEY,
    worker_name VARCHAR(50),
    client_ip VARCHAR(45) NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Konfigurasi koneksi ke Master (yaga)
CHANGE MASTER TO
    MASTER_HOST='10.78.4.9',
    MASTER_USER='repl_user',
    MASTER_PASSWORD='repl_pass',
    MASTER_LOG_FILE='mariadb-bin.000002',
    MASTER_LOG_POS=507;


START SLAVE;
SQL

# 6. Periksa status replikasi
echo "=== CEK STATUS REPLIKASI SLAVE ==="
mariadb -u root -e "SHOW SLAVE STATUS\G" | grep -E
"Slave_IO_Running:|Slave_SQL_Running:|Last_Error:"
EOF
```

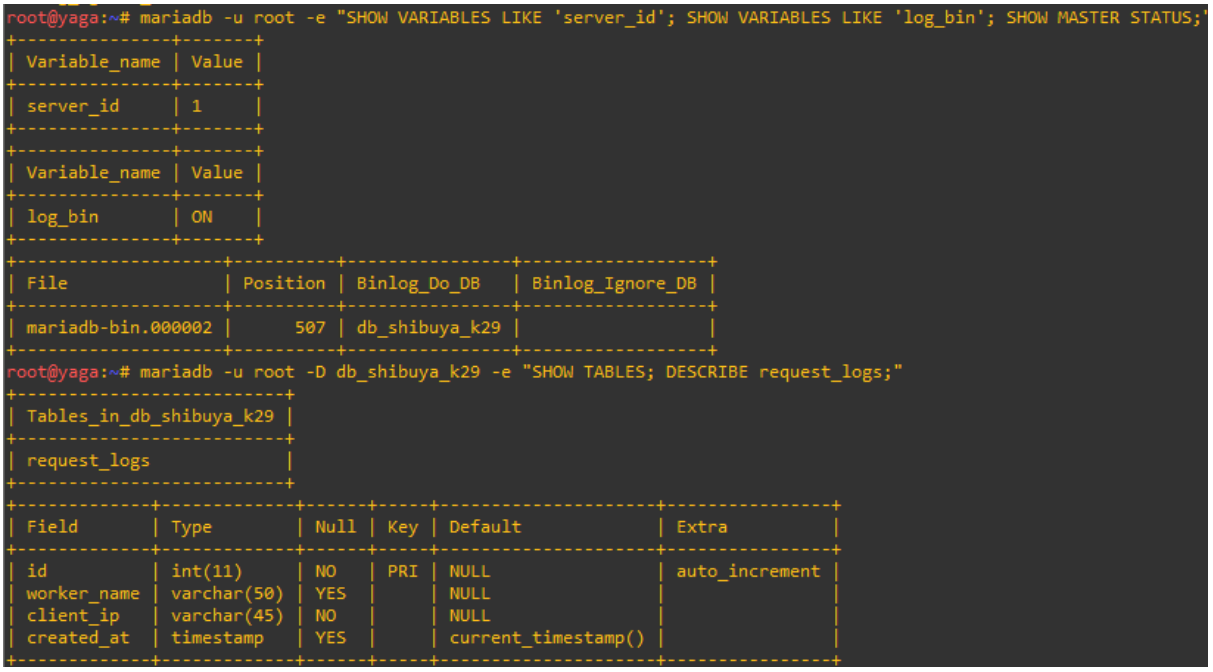
```
chmod +x /root/setup_inumaki.sh

bash /root/setup_inumaki.sh
```

1. Pembuktian config DB MASTER & Binlog di terminal yaga

```
mariadb -u root -e "SHOW VARIABLES LIKE 'server_id'; SHOW
VARIABLES LIKE 'log_bin'; SHOW MASTER STATUS;"

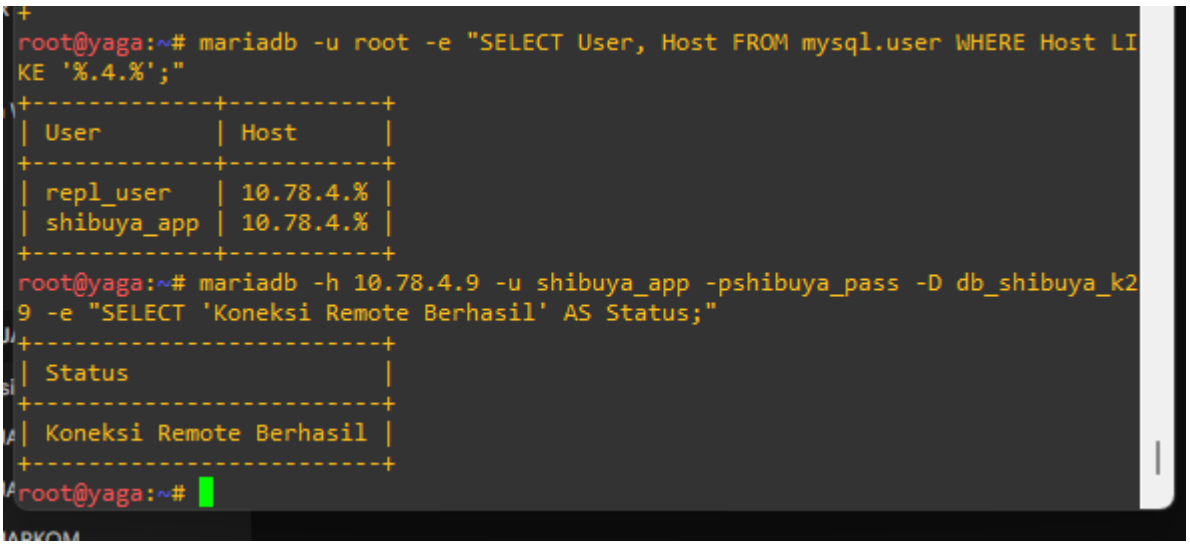
struktur database:
mariadb -u root -D db_shibuya_k29 -e "SHOW TABLES; DESCRIBE
request_logs;"
```



2. Hak akses remote network 4 YAGA

```
mariadb -u root -e "SELECT User, Host FROM mysql.user WHERE Host
LIKE '%.4.%';"

uji login langsung dari node lain di network 4
mariadb -h 10.78.4.9 -u shibuya_app -pshibuya_pass -D
db_shibuya_k29 -e "SELECT 'Koneksi Remote Berhasil' AS Status;"
```



3. Koneksi SLAVE INUMAKI

```
mariadb -u root -e "SHOW SLAVE STATUS\G"
```

4. Replikasi Data secara realtime INUMAKI

```
mariadb -u root -D db_shibuya_k29 -e "SELECT * FROM request_logs;"
*output awal kosong

mariadb -u root -D db_shibuya_k29 -e "INSERT INTO request_logs
(worker_name, client_ip) VALUES ('demo_praktikum', '10.29.2.99');"

lalu cek di yaga

mariadb -u root -e "SHOW MASTER STATUS;"

periksa kembali tabel slave
mariadb -u root -D db_shibuya_k29 -e "SELECT * FROM request_logs;"
```

SOAL 14

MAKI DAN PANDA

```
nano /root/setup_frontend.sh

#!/bin/bash

# 1. Set nameserver awal untuk instalasi paket
echo "nameserver 192.168.122.1" > /etc/resolv.conf

# 2. Instalasi Nginx dan PHP-FPM
apt-get update
apt-get install -y nginx php-fpm

# 3. Deteksi otomatis versi & socket PHP-FPM yang terpasang di
debinet
PHP_SOCK=$(find /run/php/ -name "php*-fpm.sock" | head -n 1)
if [ -z "$PHP_SOCK" ]; then
    for svc in $(service --status-all 2>&1 | grep -o 'php.*-fpm');
do
        service "$svc" start
    done
    PHP_SOCK=$(find /run/php/ -name "php*-fpm.sock" | head -n 1)
fi
echo "Menggunakan PHP Socket: $PHP_SOCK"

# 4. Siapkan direktori dan file PHP Frontend (menampilkan hostname
masing-masing)
mkdir -p /var/www/jujutsu
cat << 'PHP' > /var/www/jujutsu/index.php
<?php
header('Content-Type: text/html; charset=utf-8');
$worker = gethostname();
echo "<h1>Jujutsu High Frontend Worker</h1>";
echo "<p>Responded by: <strong>" . htmlspecialchars($worker) .
"</strong></p>";
?>
PHP
chown -R www-data:www-data /var/www/jujutsu

# 5. Konfigurasi Virtual Host & Pembatasan Akses (Hanya izinkan
dari gojo)
IP_GOJO="10.78.1.4"

cat << NGINX > /etc/nginx/sites-available/jujutsu
server {
    listen 80 default_server;
    server_name _;
    return 444;
}
```

```
server {
    listen 80;
    server_name jujutsu.k29.com www.jujutsu.k29.com;

    root /var/www/jujutsu;
    index index.php index.html;

    allow 10.78.1.4;
    allow 127.0.0.1;
    deny all;

    location / {
        try_files \$uri \$uri/ /index.php?\$query_string;
    }

    location ~ \.php\$ {
        include snippets/fastcgi-php.conf;
        fastcgi_pass unix:${PHP_SOCK};
    }
}
NGINX

# 6. Aktifkan konfigurasi dan restart service
rm -f /etc/nginx/sites-enabled/default
ln -sf /etc/nginx/sites-available/jujutsu
/etc/nginx/sites-enabled/jujutsu

PHP_SVC=$(service --status-all 2>&1 | grep -o 'php.*-fpm' | head
-n 1)
service "$PHP_SVC" restart
nginx -t && service nginx restart
chmod +x /root/setup_frontend.sh
bash /root/setup_frontend.sh
```

SKRIP SETUP WORKER API (**namai todo kirara)**

```
nano /root/setup_api_worker.sh
#!/bin/bash

# 1. Set nameserver awal untuk instalasi paket
echo "nameserver 192.168.122.1" > /etc/resolv.conf

# 2. Instalasi Nginx, PHP-FPM, dan ekstensi PHP-MySQL
apt-get update
apt-get install -y nginx php-fpm php-mysql
```

```
# 3. Deteksi otomatis socket PHP-FPM
PHP_SOCK=$(find /run/php/ -name "php*-fpm.sock" | head -n 1)
if [ -z "$PHP_SOCK" ]; then
    for svc in $(service --status-all 2>&1 | grep -o 'php.*-fpm');
do
        service "$svc" start
    done
    PHP_SOCK=$(find /run/php/ -name "php*-fpm.sock" | head -n 1)
fi
echo "Menggunakan PHP Socket: $PHP_SOCK"

# 4. Sesuaikan IP Yaga (DB Master) dan IP Geto (Internal Load
Balancer)
IP_YAGA="10.78.4.9"
IP_GETO="10.78.4.2"

# 5. Buat skrip API PHP yang mencatat log ke database yaga
mkdir -p /var/www/jujutsu
cat << PHP > /var/www/jujutsu/index.php
<?php
header('Content-Type: application/json');

\$worker_name = gethostname();
\$trusted_proxy = '{IP_GETO}';
\$remote_addr = \$_SERVER['REMOTE_ADDR'] ?? '0.0.0.0';

// Ambil IP asli klien HANYA jika trafik berasal dari proxy
terpercaya (geto)
\$client_ip = \$remote_addr;
if (\$remote_addr === \$trusted_proxy || \$remote_addr ===
'127.0.0.1') {
    if (!empty(\$_SERVER['HTTP_X_REAL_IP'])) {
        \$client_ip = trim(\$_SERVER['HTTP_X_REAL_IP']);
    } elseif (!empty(\$_SERVER['HTTP_X_FORWARDED_FOR'])) {
        \$ip_list = explode(',',
\$_SERVER['HTTP_X_FORWARDED_FOR']);
        \$client_ip = trim(\$ip_list[0]);
    }
}

// Koneksi ke MariaDB Master (yaga) sesuai kredensial Soal 13
\$db_host = '{IP_YAGA}';
\$db_user = 'shibuya_app';
\$db_pass = 'shibuya_pass';
\$db_name = 'db_shibuya_k29';

\$mysqli = @new mysqli(\$db_host, \$db_user, \$db_pass,
```

```
\$db_name);

if (\$mysqli->connect_error) {
    http_response_code(500);
    echo json_encode([
        'status' => 'error',
        'worker' => \$worker_name,
        'message' => 'DB Connection Failed: ' .
\$mysqli->connect_error
    ]);
    exit;
}

\$stmt = \$mysqli->prepare("INSERT INTO request_logs (worker_name,
client_ip) VALUES (?, ?)");
\$stmt->bind_param("ss", \$worker_name, \$client_ip);
\$stmt->execute();
\$insert_id = \$stmt->insert_id;
\$stmt->close();
\$mysqli->close();

echo json_encode([
    'status' => 'success',
    'worker' => \$worker_name,
    'client_ip' => \$client_ip,
    'log_id' => \$insert_id
]) . "\n";
?>
PHP
chown -R www-data:www-data /var/www/jujutsu

# 6. Konfigurasi Virtual Host & Pembatasan Akses (Hanya izinkan
dari geto)
cat << NGINX > /etc/nginx/sites-available/jujutsu_api
# Blokir request dengan Host header yang tidak dikenal
server {
    listen 80 default_server;
    server_name _;
    return 444;
}

# Virtual Host Layanan API
server {
    listen 80;
    server_name jujutsu.k29.com www.jujutsu.k29.com
api.jujutsu.k29.com api_cluster geto ${IP_GETO};

    root /var/www/jujutsu;
    index index.php;
```

```
    # Batasi akses: Worker API hanya menerima trafik melalui geto
    allow 10.78.4.2;
    allow 127.0.0.1;
    deny all;


    location / {
        try_files \$uri \$uri/ /index.php?\$query_string;
    }

    location ~ \.php\$ {
        include snippets/fastcgi-php.conf;
        fastcgi_pass unix:${PHP_SOCK};
    }
}
NGINX

# 7. Aktifkan konfigurasi dan restart service
rm -f /etc/nginx/sites-enabled/default
ln -sf /etc/nginx/sites-available/jujutsu_api
/etc/nginx/sites-enabled/jujutsu_api

PHP_SVC=$(service --status-all 2>&1 | grep -o 'php.*-fpm' | head
-n 1)
service "\$PHP_SVC" restart
nginx -t && service nginx restart


chmod +x /root/setup_api_worker.sh && bash
/root/setup_api_worker.sh
```

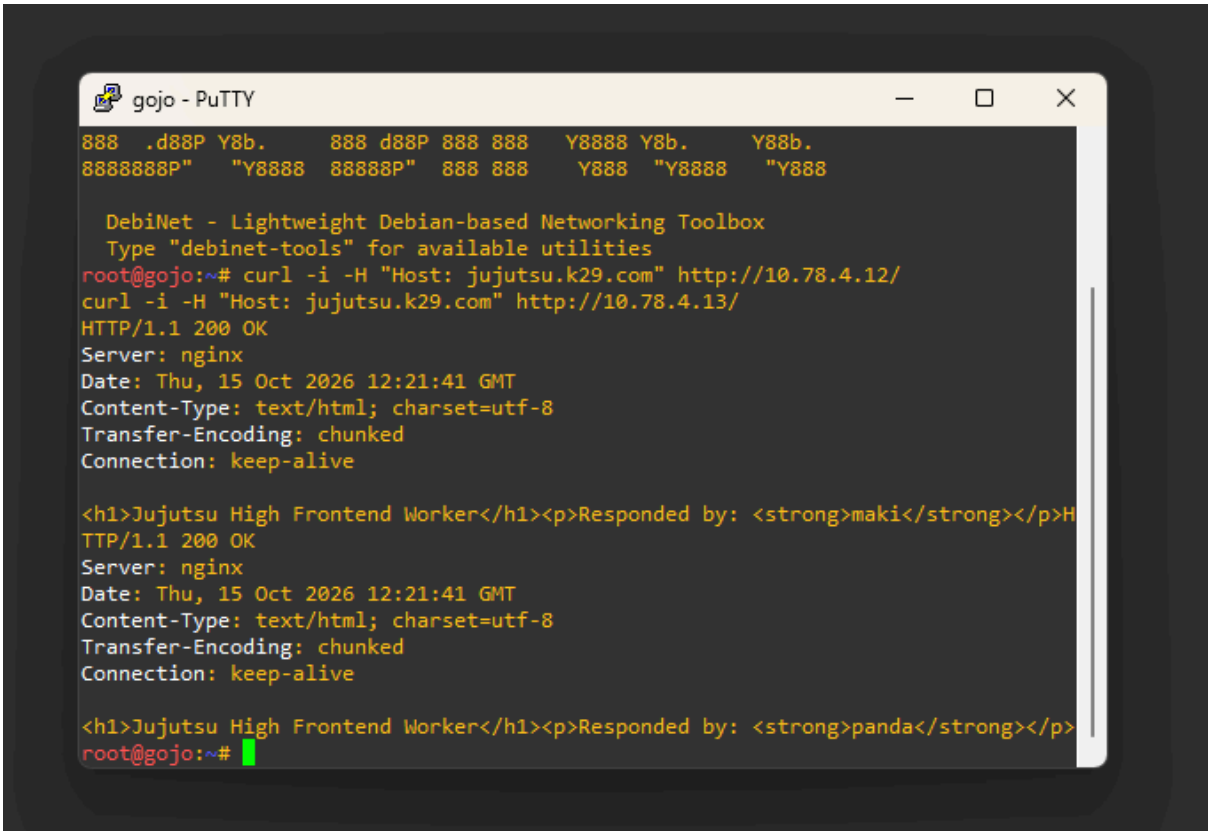
1. UJI AKSES LANGSUNG DARI KLIEN (HARUS DITOLAK) itadori

```
curl -i -H "Host: jujutsu.k29.com" http://10.78.4.12/
curl -i -H "Host: jujutsu.k29.com" http://10.78.4.14/api
```

2. UJI AKSES MELALUI JALUR PROXY

Dari terminal gojo, uji koneksi ke maki dan panda

```
curl -i -H "Host: jujutsu.k29.com" http://10.78.4.12/
curl -i -H "Host: jujutsu.k29.com" http://10.78.4.13/
```



terminal geto, uji konseki ke nanami, todo dan kirara

```
curl -i -H "Host: jujutsu.k29.com" http://10.78.4.14/api
curl -i -H "Host: jujutsu.k29.com" http://10.78.4.15/api
curl -i -H "Host: jujutsu.k29.com" http://10.78.4.16/api
```

```
root@geto:~# curl -i -H "Host: jujutsu.k29.com" http://10.78.4.14/api
curl -i -H "Host: jujutsu.k29.com" http://10.78.4.15/api
curl -i -H "Host: jujutsu.k29.com" http://10.78.4.16/api
HTTP/1.1 200 OK
Server: nginx
Date: Thu, 15 Oct 2026 01:46:44 GMT
Content-Type: application/json
Transfer-Encoding: chunked
Connection: keep-alive

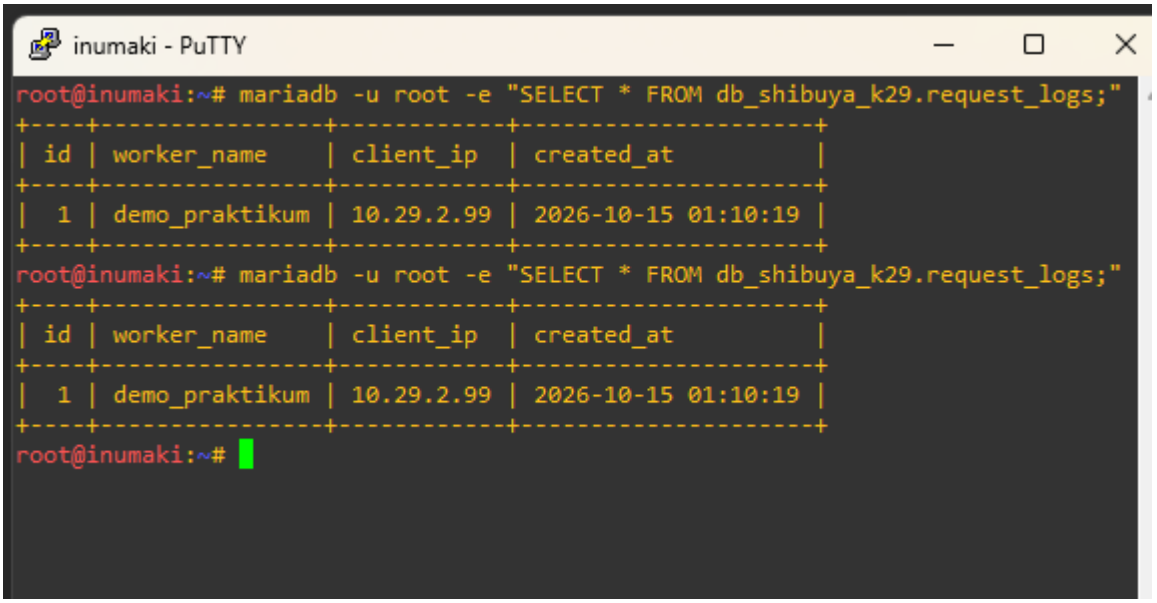
{"status":"success","worker":"nanami","client_ip":"10.78.4.2","log_id":6}
HTTP/1.1 200 OK
Server: nginx
Date: Thu, 15 Oct 2026 01:46:44 GMT
Content-Type: application/json
Transfer-Encoding: chunked
Connection: keep-alive

{"status":"success","worker":"todo","client_ip":"10.78.4.2","log_id":7}
HTTP/1.1 200 OK
Server: nginx
Date: Thu, 15 Oct 2026 01:46:44 GMT
Content-Type: application/json
Transfer-Encoding: chunked
Connection: keep-alive

{"status":"success","worker":"kirara","client_ip":"10.78.4.2","log_id":8}
root@geto:~#
```

3. Pencatatan log di database yaga dan replikasi di inumaki

`mariadb -u root -e "SELECT * FROM db_shibuya_k29.request_logs;"`



SOAL 15

```
cat << 'EOF' > /root/setup.sh
#!/bin/bash
```

```
echo "nameserver 192.168.122.1" > /etc/resolv.conf

apt-get update
apt-get install -y nginx

cat << 'NGINX' > /etc/nginx/sites-available/default
upstream api_cluster {
    server 10.78.4.14:80 weight=3; # nanami
    server 10.78.4.15:80 weight=1; # todo
}

server {
    listen 80 default_server;
    listen [::]:80 default_server;
    server_name _;

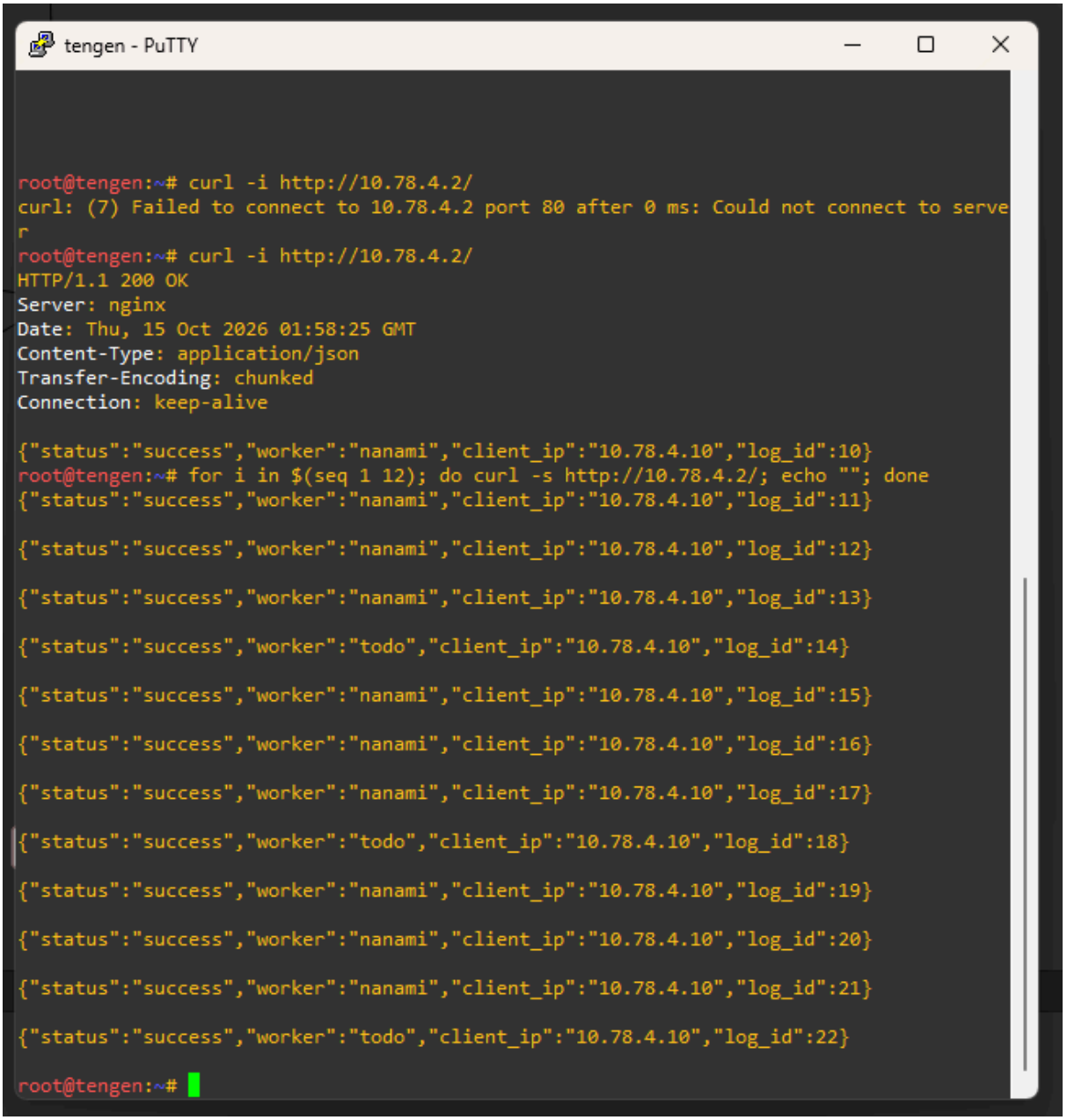
    location / {
        proxy_pass http://api_cluster;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    }
}
NGINX

ln -sf /etc/nginx/sites-available/default
/etc/nginx/sites-enabled/default
nginx -t && service nginx restart
EOF

chmod +x /root/setup.sh
bash /root/setup.sh



tengen
curl -i http://10.78.4.2/
for i in $(seq 1 12); do curl -s http://10.78.4.2/; echo ""; done
```



SOAL 16
nano /root/setup_gojo16.sh

SOAL 17

/root/config_gojo.sh

```
#!/bin/bash

# Sesuaikan variabel berikut dengan Master Sheet kelompok Anda
```

```
PREFIX_IP="10.x"          # Ganti dengan 2 oktet PREFIX IP kelompok
KELOMPOK="k01"            # Ganti dengan kode kelompok <xxxx>
IP_GETO="${PREFIX_IP}.4.x" # Ganti dengan IP statis geto di Network
4
IP_MAKI="${PREFIX_IP}.4.x" # Ganti dengan IP statis maki di Network
4
IP_PANDA="${PREFIX_IP}.4.x" # Ganti dengan IP statis panda di
Network 4

cat <<EOF > /etc/nginx/sites-available/jujutsu
upstream frontend_cluster {
    server ${IP_MAKI};
    server ${IP_PANDA};
}

server {
    listen 80;
    server_name jujutsu.KELOMPOK.comwww.jujutsu.{KELOMPOK}.com;

    location / {
        proxy_pass http://frontend_cluster;
        proxy_set_header Host \$host;
        proxy_set_header X-Real-IP \$remote_addr;
        proxy_set_header X-Forwarded-For
\$proxy_add_x_forwarded_for;
    }

    location /api {
        proxy_pass http://${IP_GETO};
        proxy_set_header Host \$host;
        # Meneruskan IP asli klien yang terhubung ke gojo
        proxy_set_header X-Real-IP \$remote_addr;
        proxy_set_header X-Forwarded-For
\$proxy_add_x_forwarded_for;
    }
}
EOF

ln -sf /etc/nginx/sites-available/jujutsu
/etc/nginx/sites-enabled/jujutsu
rm -f /etc/nginx/sites-enabled/default
nginx -t && service nginx reload
```

/root/config_geto.sh

```
#!/bin/bash

PREFIX_IP="10.x"            # Ganti dengan 2 oktet PREFIX IP
kelompok
IP_GOJO="${PREFIX_IP}.1.x"   # Ganti dengan IP statis gojo di
Network 1
IP_NANAMI="${PREFIX_IP}.4.x" # Ganti dengan IP statis nanami
IP_TODO="${PREFIX_IP}.4.x"   # Ganti dengan IP statis todo

cat <<EOF > /etc/nginx/sites-available/api_lb
upstream api_cluster {
    # Sesuai Soal 15: Weighted Round Robin 3:1 untuk nanami dan todo
    server ${IP_NANAMI} weight=3;
    server ${IP_TODO} weight=1;
}

server {
    listen 80;
    server_name _;

    # Mempercayai HANYA proxy gojo untuk ekstraksi IP asli klien
    set_real_ip_from ${IP_GOJO};
    real_ip_header X-Real-IP;
    real_ip_recursive on;

    location / {
        proxy_pass http://api_cluster;
        proxy_set_header Host \$host;
        # Karena real_ip_header aktif, \$remote_addr di sini sudah
berisi IP asli klien
        proxy_set_header X-Real-IP \$remote_addr;
        proxy_set_header X-Forwarded-For
\$proxy_add_x_forwarded_for;
    }
}
EOF

ln -sf /etc/nginx/sites-available/api_lb
/etc/nginx/sites-enabled/api_lb
rm -f /etc/nginx/sites-enabled/default
nginx -t && service nginx reload

nanami todo kirara
```

```
server {
    listen 80;
    server_name jujutsu.<xxxx>.com www.jujutsu.<xxxx>.com _;

    root /var/www/html;
    index index.php;

    # Batasi akses hanya dari geto (Soal 14)
    allow <IP_GETO>;
    deny all;

    # Hanya percaya header IP klien yang dikirim oleh geto (Soal 17)
    set_real_ip_from <IP_GETO>;
    real_ip_header X-Real-IP;
    real_ip_recursive on;

    location /api {
        try_files $uri uri//api/index.php?query_string;
    }

    location ~ \.php$ {
        include snippets/fastcgi-php.conf;
        fastcgi_pass unix:/run/php/php-fpm.sock; # Sesuaikan versi
PHP di container, misal php8.2-fpm.sock
        fastcgi_param SCRIPT_FILENAME document_rootfastcgi_script_name;
        include fastcgi_params;
    }
}

index.php pada nanami todo dan kirara
<?php
header('Content-Type: application/json');

$db_host = '<IP_YAGA>'; // Ganti dengan IP statis yaga di Network 4
$db_user = '<DB_USER>'; // User database yang diberi akses remote di
Soal 13
$db_pass = '<DB_PASS>'; // Password database
$db_name = 'db_shibuya_<xxxx>'; // Sesuai Soal 13

// Karena Nginx worker sudah membatasi allow hanya dari geto dan
memasang set_real_ip_from geto,
// maka REMOTE_ADDR adalah IP klien yang telah diverifikasi.
$client_ip = $_SERVER['REMOTE_ADDR'];
```

```
$x_forwarded_for = $_SERVER['HTTP_X_FORWARDED_FOR'] ?? $client_ip;
$hostname = gethostname();

// Validasi format IP sebelum disimpan
if (!filter_var($client_ip, FILTER_VALIDATE_IP)) {
    http_response_code(400);
    echo json_encode(["status" => "error", "message" => "Invalid
Client IP"]);
    exit;
}

mysqli = newmysqli(db_host, $db_user, $db_pass, $db_name);

if ($mysqli->connect_error) {
    http_response_code(500);
    echo json_encode(["status" => "error", "message" => "DB
Connection Failed: " . $mysqli->connect_error]);
    exit;
}

// Menyimpan client_ip ke tabel request_logs sesuai Soal 17
$stmt = $mysqli->prepare("INSERT INTO request_logs (client_ip)
VALUES (?)");
$stmt->bind_param("s", $client_ip);
$stmt->execute();
$stmt->close();
$mysqli->close();

echo json_encode([
    "status" => "success",
    "handled_by" => $hostname,
    "verified_client_ip" => $client_ip,
    "x_forwarded_for" => $x_forwarded_for
]);
?>


uji klien network 2
# Cek IP klien saat ini
ip a show eth0
# Kirim request ke endpoint /api
curl -i http://jujutsu.<xxxx>.com/api

network 3
```

```
# Cek IP klien saat ini
ip a show eth0
# Kirim request ke endpoint /api
curl -i http://jujutsu.<xxxx>.com/api

verif requestlog pada yaga
mysql -u root -p -e "USE db_shibuya_<xxxx>; SELECT * FROM
request_logs ORDER BY id DESC LIMIT 10;"
```

SOAL 18
SOAL 19
SOAL 20